Sentiment Analysis of IMDb Movie Reviews

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DS 4002 - Data Science Project

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Motivation:

- Classify movie reviews to **determine sentiment**
- Evaluate whether review sentiment **predict IMDb ratings**
- Applications:
  - Recommendation systems
  - Audience insight analysis

Hypothesis: Higher sentiment scores are associated with higher rating categories

Research question: Is there a strong correlation between IMDb review and the corresponding movie rating?

Modeling approach:

- Sentiment analysis with python VADER package
- Linear regression with sentiment vs movie rating

Goal: Investigate whether movie review sentiment correlate strongly enough with rating to be used as a predictor

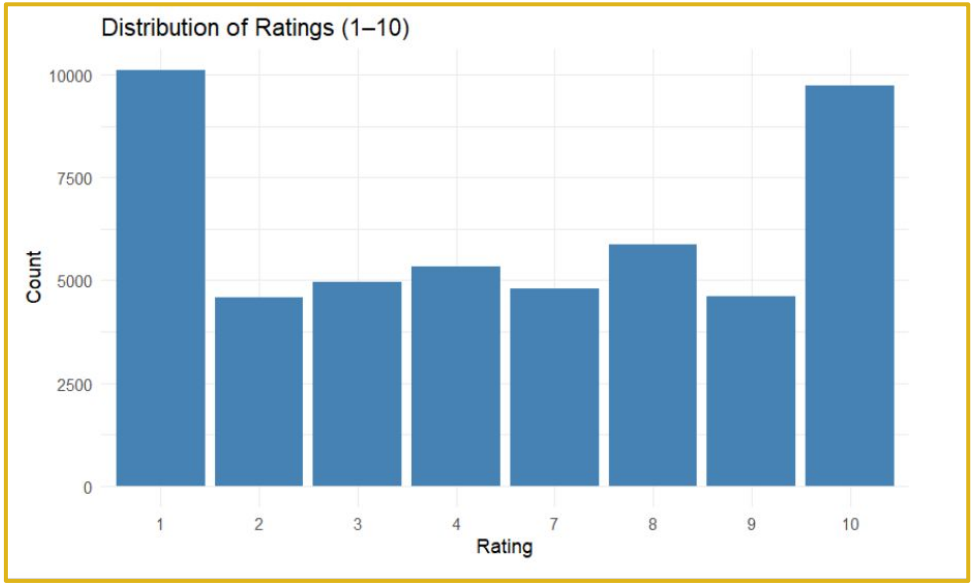
Data Acquisition and Explanation

Column	Description
Review	Text based evaluation
Rating	Numerical rating the user has assigned the movie only includes ratings 1-4, and 7-10
Sentiment Score	Binary score based on the rating. (replaces the old sentiment score column)
Compound	Any value from -1 to 1. Represents the strength of a sentiment

Summary of original dataset:

- Dataset had 3 columns:
  - Review
  - Rating
  - Sentiment (0,1)
- 50,000 rows of data
- Origin: Preprocessed Stanford IMDb movie review dataset from Kaggle

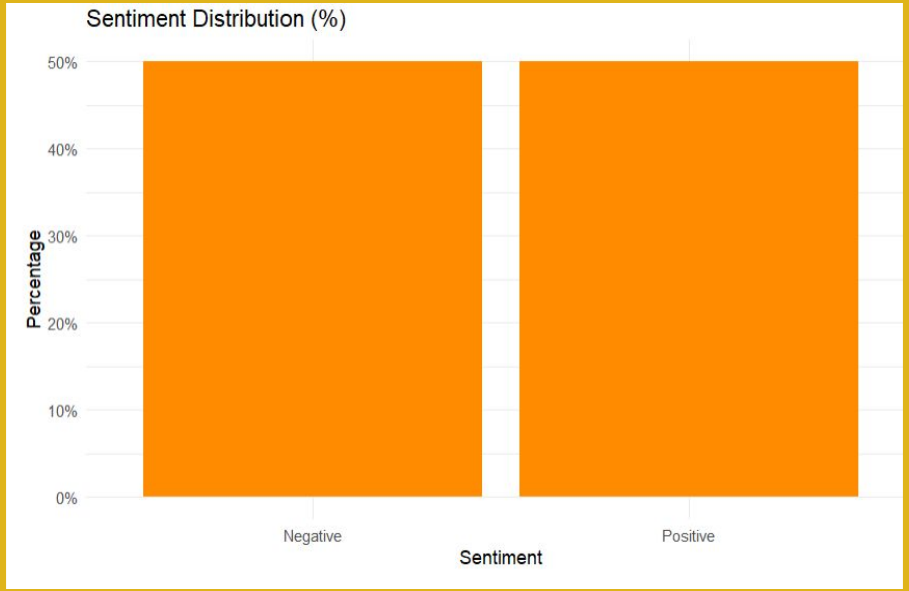
Analysis Plan

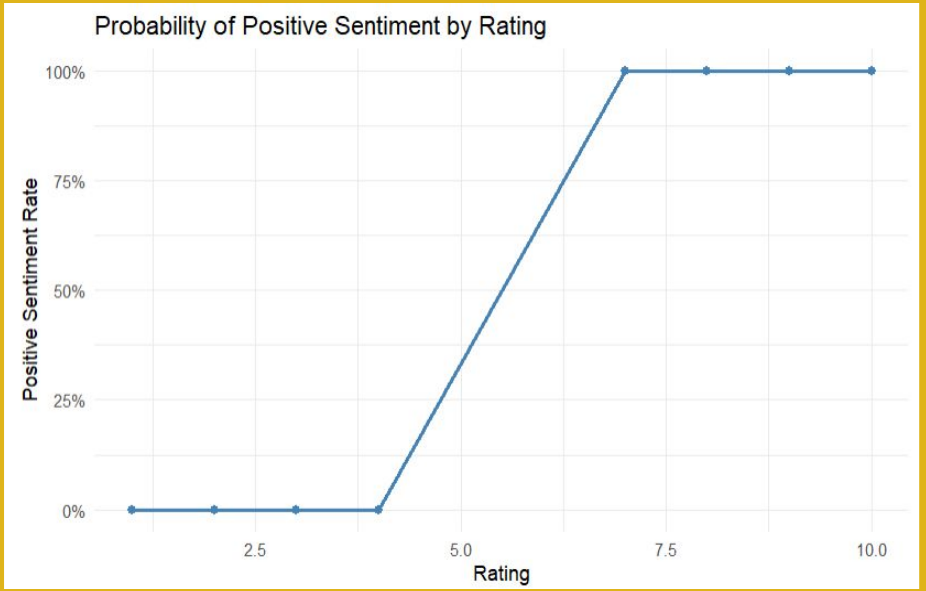


Steps:

1. **Load IMDb dataset and do EDA**
2. **Sentiment Analysis with VADER**
   - compute compound score pre review
3. **Feature Engineering**
   - Keep compound as predictor
4. **Model**
   - Ordered logistic regression
5. **Predictions**
   - Probabilities per rating
6. **Evaluation**
   - MAE, R^2

Challenge Faced: Using Ordinal Regression


Sentiment Distribution (%)
50%
40%
30%
20%
10%
0%
Percentage
Negative
Positive
Sentiment


Probability of Positive Sentiment by Rating
100%
75%
50%
25%
0%
Positive Sentiment Rate
2.5
5.0
7.5
10.0
Rating

Tricky Analysis Decision

Limitations of Linear Regression

- Treats ratings as continuous intervals
- Wrongly assumes the "distance" between every rating (e.g., 1-2 vs. 9-10) is identical

Advantage of Ordinal Linear Regression

- Treats ratings as ordered groups
- Jump from 1-2 may differ from the jump from 7-8

H_0: The VADER compound sentiment score has no effect on the probability of a movie being classified into a higher rating category.

H_a: As the VADER compound sentiment score increases, the log-odds of the movie review falling into a higher rating category increase significantly.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 Rating   R-squared:                       0.226
Model:                            OLS   Adj. R-squared:                  0.226
Method:                 Least Squares   F-statistic:                 1.459e+04
Date:                Wed, 11 Feb 2026   Prob (F-statistic):               0.00
Time:                        15:50:38   Log-Likelihood:            -1.2688e+05
No. Observations:               50000   AIC:                         2.538e+05
Df Residuals:                   49998   BIC:                         2.538e+05
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          4.8574      0.015    331.096      0.000       4.829       4.886
compound       2.0555      0.017    120.808      0.000       2.022       2.089
==============================================================================
Omnibus:                     4279.692   Durbin-Watson:                   0.639
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1371.063
Skew:                          -0.069   Prob(JB):                    1.89e-298
Kurtosis:                       2.200   Cond. No.                         1.50
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly

Linear MAE: 2.592
Linear R2: 0.226
```

```
                           OrderedModel Results
==============================================================================
Dep. Variable:                 Rating   Log-Likelihood:                -95353.
Model:                   OrderedModel   AIC:                         1.907e+05
Method:             Maximum Likelihood   BIC:                         1.908e+05
Date:                Mon, 09 Feb 2026
Time:                        15:27:41
No. Observations:               50000
Df Residuals:                   49992
Df Model:                           1
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
compound       1.1295      0.011    104.176      0.000       1.108       1.151
1/2           -1.2720      0.012   -107.606      0.000      -1.295      -1.249
2/3           -0.5440      0.014    -38.510      0.000      -0.572      -0.516
3/4           -0.6234      0.014    -45.880      0.000      -0.650      -0.597
4/7           -0.6188      0.013    -47.270      0.000      -0.644      -0.593
7/8           -0.7447      0.014    -53.980      0.000      -0.772      -0.718
8/9           -0.5097      0.012    -41.328      0.000      -0.534      -0.486
9/10          -0.5859      0.014    -41.661      0.000      -0.613      -0.558
==============================================================================

--- Evaluation ---
Mean Absolute Error (MAE): 2.606
R-squared (R2): 0.227
```

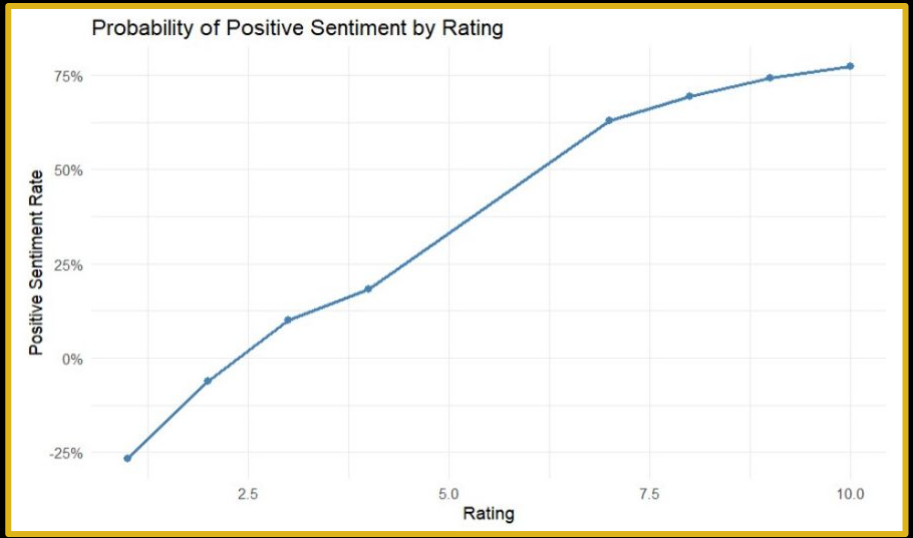
Bias and Uncertainty Validation

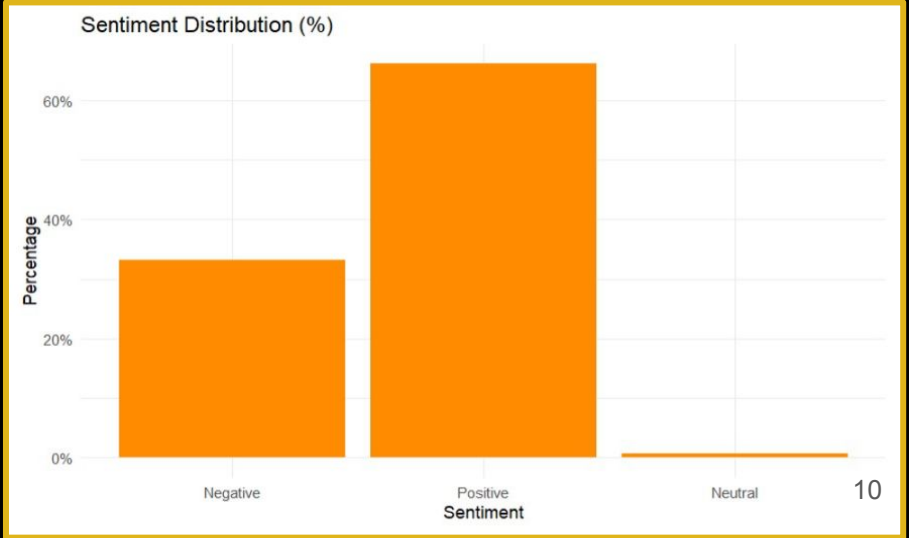
- No movie reviews with ratings of 5-6
- "Bad" R^2 (0.227) and MAE(2.606) could be due to:
  - Sarcasm
  - Current mood of critic
  - Personal standards

Score	Sentiment	Compound Score
7	-1	-0.4574

Results and Conclusions

- VADER compound sentiment is a strong and statistically significant predictor of rating
- As sentiment increases, the probability of higher rating categories increases
- Sentiment explains about 23% of rating variation (R^2 = 0.227)
  - Other factors influence ratings as well





Next Steps

New Lines of Exploration:

- Compare sentiment effects across features (genre, movie length)
- Analyze prediction differences between extreme and mid range ratings

improvements:

- Add a train/test validation
- Improve handling for sarcasm and emojis

New Questions:

- Are extreme ratings easier to predict?
- Does punctuation correlated to extreme ratings?
- How has sentiment scores changed over time?

References

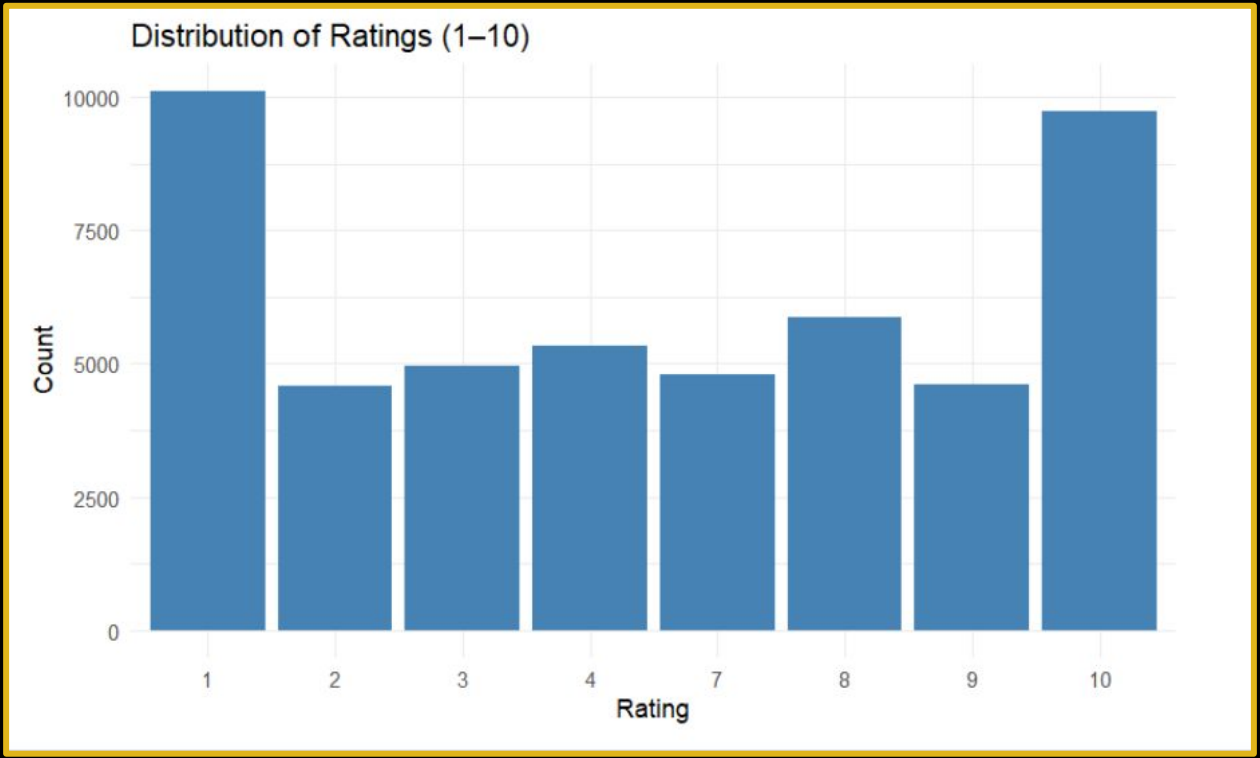
[1] P. Frangidis, K. Georgiou, and S. Papadopoulos,
“Sentiment Analysis on Movie Scripts and Reviews: Utilizing Sentiment Scores in Rating Prediction,”
in Artificial Intelligence Applications and Innovations, 2020.
[Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC7256373/

[2] Ren, X., & Chandra, R. (2025). *IMDb movie reviews analysis using a deep learning framework: Rating prediction, sentiment analysis, and rating-tier analysis*. arXiv. https://arxiv.org/abs/2601.07235

[3] C. J. Hutto and E. Gilbert, “Welcome to VADER Sentiment’s documentation,” VADER Sentiment 3.3.1 Documentation. [Online]. Available: https://vadersentiment.readthedocs.io/en/latest/pages/introduction.html

Github: https://github.com/isaiahluc/M3MovieSentiment/tree/main

Questions?


Distribution of Ratings (1–10)
10000
7500
5000
2500
0
Count
1
2
3
4
7
8
9
10
Rating